

$|a|$ is the distance between a and 0 on the real number line.

$$|a| \geq 0 \text{ for all } a \in \mathbb{R}$$

The definition of the absolute value:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

This means: $|-3| = 3$ $|27| = 27$

$$|-a| = |a|$$

$$|-x^2| = x^2$$

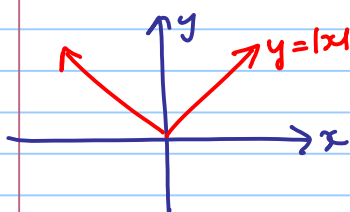
Obj: Know the properties of the absolute value on p. A7.

We use the definition to graph absolute value functions:

Graph:

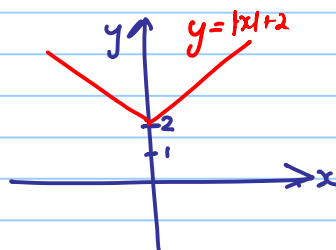
$$\textcircled{1} y = |x|$$

$$= \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



$$\textcircled{2} y = |x| + 2$$

$$= \begin{cases} x + 2 & \text{if } x \geq 0 \\ -x + 2 & \text{if } x < 0 \end{cases}$$



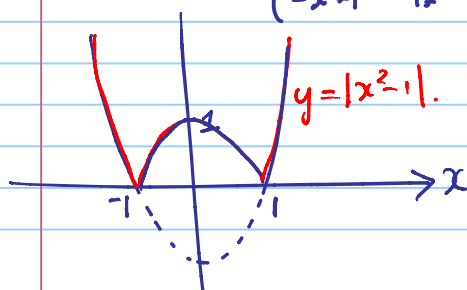
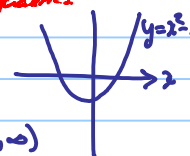
Also: $|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$

Example: Sketch the graph of $y = |x^2 - 1|$.

$$y = |x^2 - 1| = \begin{cases} x^2 - 1 & \text{if } x^2 - 1 \geq 0 \\ -(x^2 - 1) & \text{if } x^2 - 1 < 0 \end{cases}$$

$$= \begin{cases} x^2 - 1 & \text{if } x \in (-\infty, -1] \cup [1, \infty) \\ -x^2 + 1 & \text{if } x \in (-1, 1) \end{cases}$$

From Unit 1.1: We use a graph to solve these inequalities.



Theorem: (see p. A7: (b))

Suppose $a > 0$. Then

$$1) |x| = a \text{ if and only if } x = \pm a.$$

$$2) |x| < a \text{ if and only if } -a < x < a.$$

$$3) |x| > a \text{ if and only if } x > a \text{ or } x < -a.$$

[To solve inequalities, we use this Theorem.

Examples.

① See Examples 6, 7 and 8 on pp. A7-A8.

$$② |x^2 - 9| = 1$$

$$\Rightarrow x^2 - 9 = 1$$

$$\Rightarrow x^2 = 10$$

$$\Rightarrow x = \pm \sqrt{10}$$

or

$$x^2 - 9 = -1$$

$$x^2 = -1 + 9 = 8$$

$$x = \pm \sqrt{8}$$

$$\Rightarrow x \in \{-\sqrt{10}, -\sqrt{8}, \sqrt{8}, \sqrt{10}\}.$$

$$3) |x+1| = -1$$

$$\Rightarrow x \in \emptyset \quad (\text{WHY??})$$

$$4) |x-4| < -4$$

$$\Rightarrow x \in \emptyset$$

$$|x-4| > -4$$

$$\Rightarrow x \in \mathbb{R}.$$

$$5) \left| \frac{2x-1}{x+1} \right| = 3$$

$$\Rightarrow \frac{2x-1}{x+1} = \pm 3$$

$$\Rightarrow \frac{2x-1}{x+1} = 3$$

or

$$\frac{2x-1}{x+1} = -3$$

$$\Rightarrow 2x-1 = 3(x+1)$$

$$\Rightarrow 2x-1 = 3x+3$$

$$\Rightarrow -x = 4$$

$$\Rightarrow x = -4$$

$$2x-1 = -3(x+1) = -3x-3$$

$$\Rightarrow 5x = -3+1 = -2$$

$$\Rightarrow x = -\frac{2}{5}$$

$$\Rightarrow x = -4 \text{ or } x = -\frac{2}{5}.$$

$$(6) |x+3| = |2x+1|$$

This equation we solve by squaring both sides, on which case you always have to test your answer!!

$$\begin{aligned} |x+3|^2 &= |2x+1|^2 & (2x+1)^2 &= (2x+1)(2x+1) \\ & & &= 4x^2 + 4x + 1 \\ \Rightarrow x^2 + 6x + 9 &= 4x^2 + 4x + 1 \\ \Rightarrow -3x^2 + 2x + 8 &= 0 \\ \Rightarrow 3x^2 - 2x - 8 &= 0 \\ \Rightarrow (3x+4)(x-2) &= 0 \\ \Rightarrow 3x+4=0 &\text{ or } x-2=0 \\ \Rightarrow x=-\frac{4}{3} &\text{ or } x=2. \end{aligned}$$

$$\begin{aligned} \text{Test: If } x = -\frac{4}{3}: \text{ LHS} &= |x+3| = |-\frac{4}{3}+3| = |\frac{5}{3}| = \frac{5}{3} \\ \text{RHS} &= |2x+1| = |2(-\frac{4}{3})+1| = |-\frac{8}{3}+1| = |-\frac{5}{3}| = \frac{5}{3} \end{aligned}$$

$$\begin{aligned} \text{If } x = 2: \text{ LHS} &= |x+3| = |2+3| = 5 \\ \text{RHS} &= |2x+1| = |2(2)+1| = 5. \end{aligned}$$

$$(7) \frac{x}{|x-2|} > 0$$

$$\Rightarrow x > 0 \text{ and } x \neq 2$$

$$\Rightarrow x \in (0, 2) \cup (2, \infty).$$

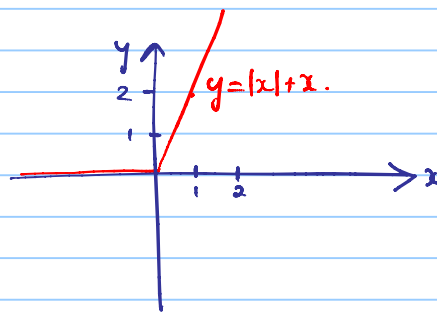
$$(8) \left| \frac{x+1}{x-3} \right| > 0$$

$$x \neq -1, x \neq 3 \Rightarrow x \in \mathbb{R} \setminus \{-1, 3\}$$

$$(9) \text{Sketch } y = |x| + 2$$

Sketch: Use definition!!

$$\begin{aligned} y &= |x| + 2 \\ &= \begin{cases} x+2 & \text{if } x \geq 0 \\ -x+2 & \text{if } x < 0 \end{cases} \\ &= \begin{cases} 2x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \end{aligned}$$



Some remarks:

$$1) \sqrt{x^2} = |x|, \text{ e.g. } \sqrt{25} = 5.$$

$$\text{but: } x^2 = 25 \Rightarrow x = \pm 5.$$

2) To solve an equation: use property 4 on p. A7

To solve an inequality: use properties 5, 6 on p. A7.

To sketch a graph: use the definition.